

MAP 112 - Álgebra Linear

Sistemas Lineares

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## Exercício de Direção

\* Resolva e classifique os sistemas homogêneos abaixo.

1. (3x3)

$$\begin{cases} x + y + z = 0 \\ x - y + z = 0 \\ 2x + 2y + 2z = 0 \end{cases}$$

|   |    |   |   |
|---|----|---|---|
| 1 | 1  | 1 | 0 |
| 1 | -1 | 1 | 0 |
| 2 | 2  | 2 | 0 |

$$\begin{matrix} 1 & L \\ 3 & 3 \\ m & m \end{matrix}$$

Matriz aumentada

L2 -&gt; L2 - L1

$$= 1 - 1 = 0$$

$$= -1 - 1 = -2$$

$$= 1 - 1 = 0$$

$$= 0 - 0 = 0$$

L3 -&gt; L3 - 2L1

$$= 2 - 2 \cdot 1 = 2 - 2 = 0$$

$$= 2 - 2 \cdot 1 = 2 - 2 = 0$$

$$= 2 - 2 \cdot 1 = 2 - 2 = 0$$

$$= 0 - 2 \cdot 0 = 0$$

| x | y  | z |   |
|---|----|---|---|
| 1 | 1  | 1 | 0 |
| 0 | -2 | 0 | 0 |
| 0 | 0  | 0 | 0 |

Rango  $r(A) = 2$ Incógnitas  $m = 3$ Número de Parâmetros  $= 3 - 2 = 1$ 

Sistema Possível Indeterminado SPI

$$\begin{cases} x + y + z = 0 \\ 2y = 0 \end{cases}$$

Segunda equação (linha)

$$-2y = 0$$

$$y = \frac{0}{2} = 0$$

Primeira equação (linha)

$$x + y + z = 0$$

$$x + 0 + z = 0$$

$$x = -z$$

$z = t$  porque a coluna não tem pivô

$$(x, y, z) = (-t, 0, t)$$

$$t(-1, 0, 1)$$

2 (3x3)

$$\begin{cases} x + y + z = 0 \\ 2x + y + z = 0 \\ 3x + 2y + 3z = 0 \end{cases}$$

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 1 | 0 |
| 2 | 1 | 1 | 0 |
| 3 | 3 | 2 | 0 |

$$L2 \rightarrow L2 - L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 0 - 2 \cdot 0 = 0$$

$$L3 \rightarrow L3 - L1$$

$$= 1 - 1 = 0$$

$$= 2 - 1 = 1$$

$$= 3 - 1 = 2$$

$$= 0 - 0 = 0$$

|   |    |    |   |
|---|----|----|---|
| 1 | 1  | 1  | 0 |
| 0 | -1 | -1 | 0 |
| 0 | 1  | 2  | 0 |

$$L3 \rightarrow L3 + L2$$

$$= 1 + (-1) = 0$$

$$= 2 + (-1) = 1$$

$$= 0 + 0 = 0$$

|   |    |    |   |
|---|----|----|---|
| 1 | 1  | 1  | 0 |
| 0 | -1 | -1 | 0 |
| 0 | 0  | 1  | 0 |

tilibra

$$\text{Rote } r(A) = 3$$

$$\text{Inequilíbrios} = 3$$

$$\text{Número de Parâmetros} = 3 - 3 = 0$$

Sistema Possível Determinado SPD

$$\begin{cases} x + y + z = 0 \\ -y - z = 0 \\ -z = 0 \end{cases}$$

Terceira Linha

$$-z = 0$$

Segunda Linha

$$-y - z = 0$$

$$-y - 0 = 0$$

$$-y = 0$$

Terceira Linha

$$x + y + z = 0$$

$$x + 0 + 0 = 0$$

$$x = 0$$

$$(x, y, z) = (0, 0, 0)$$

$$3 \times (3 \times 4)$$

$$\begin{cases} x + y + z + t = 0 \\ 2x + y + z + t = 0 \\ x - y + z + t = 0 \end{cases}$$

$$\begin{matrix} 1 & 2 \\ 4 & 3 \end{matrix}$$

$$m \quad m$$

|   |   |    |   |   |
|---|---|----|---|---|
| 1 | 1 | 1  | 1 | 0 |
| 2 | 1 | 1  | 1 | 0 |
| 3 | 1 | -1 | 1 | 0 |

$$L2 = L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 0 - 2 \cdot 1 = 0$$

$$L3 = L3 - L1$$

$$= 1 - 1 = 0$$

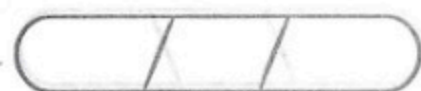
$$= -1 - 1 = -2$$

$$= 1 - 1 = 0$$

$$= 1 - 1 = 0$$

$$= 0 - 0 = 0$$

|   |    |    |    |   |
|---|----|----|----|---|
| 1 | 1  | 1  | 1  | 0 |
| 0 | -1 | -1 | -1 | 0 |
| 0 | -2 | 0  | 0  | 0 |



|                             | x | y | z  | t  |   |
|-----------------------------|---|---|----|----|---|
| $13^o \quad 13+212$         | ① | 1 | 1  | 1  | 0 |
| $= -2 - 2(-1) = -2 + 2 = 0$ | 0 | ① | -1 | -1 | 0 |
| $= 0 - 2(-1) = 0 + 2 = 2$   | 0 | 0 | ②  | 2  | 0 |
| $= 0 - 2(-1) = 0 + 2 = 2$   |   |   |    |    |   |
| $= 0 - 2 \cdot 0 = 0 = 0$   |   |   |    |    |   |

Rato  $n(A) = 3$

Inequilios  $m = 4$

Número de Parâmetros  $= 4 - 3 = 1$

Sistema Possível Indeterminado SPI

$$\begin{cases} x + y + z + t = 0 \\ -y - z - t = 0 \\ 2y + 2t = 0 \end{cases}$$

Terceira Linha

$$2y + 2t = 0$$

$$2y + 2w = 0$$

$$2y = -2w$$

$$y = -w$$

~~z~~

$$y = -w$$

Segunda Linha

$$-y - z - t = 0$$

$$-y - (-w) - w = 0$$

$$-y + w - w = 0$$

$$-y = 0 + w - w = 0$$

$$-y = 0$$

Primeira Linha

$$x + y + z + t = 0$$

$$x + 0 + (-w) + w = 0$$

$$x - w + w = 0$$

$$x = 0$$

$$(x, y, z, t) = (0, 0, -w, w)$$

$$= w(0, 0, -1, 1)$$

2. (2x4)

$$\begin{cases} x + y + z + t = 0 \\ 2x + 2y + 2z + 2t = 0 \end{cases}$$

1 L  
4 J 2  
m m

tilibra

|   |   |   |   |   |
|---|---|---|---|---|
| ① | 1 | 1 | 1 | 0 |
| 2 | 2 | 2 | 2 | 0 |

$$12 = 12 - 2 \cdot 1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 0 - 2 \cdot 0 = 0$$

|   | x | y | z | t |
|---|---|---|---|---|
| ① | 1 | 1 | 1 | 0 |
|   | 0 | 0 | 0 | 0 |

$$\text{Rango } m(A) = 1$$

$$\text{Incógnitas } m = 4$$

$$\text{Número de Parámetros} = 4 - 1 = 3$$

$$y = 1a$$

$$z = 1b$$

$$t = 1c$$

Sistema ~~Resuelto~~ Indeterminado SPI

$$\begin{cases} x + y + z + t = 0 \end{cases}$$

$$x + 1a + 1b + 1c = 0$$

$$x = 0 - 1a - 1b - 1c$$

$$(x, y, z, t) = (-1a - 1b - 1c, 1a, 1b, 1c)$$

$$= 1a(-1, 1, 0, 0) + 1b(-1, 0, 1, 0) + 1c(-1, 0, 0, 1)$$

5. (4x3)

$$\begin{cases} x + y + z = 0 \\ 2x + 2y + 2z = 0 \\ x - y + z = 0 \\ 3x + y + z = 0 \end{cases}$$

$$\begin{matrix} 1 & 2 \\ 3 & 4 \\ m & m \end{matrix}$$

|   |    |   |   |
|---|----|---|---|
| ① | 1  | 1 | 0 |
| 2 | 2  | 2 | 0 |
| 1 | -1 | 1 | 0 |
| 3 | 1  | 1 | 0 |

tilibra

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 0 - 2 \cdot 0 = 0$$

$$L3 \rightarrow L3 - L1$$

$$= 1 - 1 = 0$$

$$= -1 - 1 = -2$$

$$= 1 - 1 = 0$$

$$= 0 - 0 = 0$$

$$L4 \rightarrow L4 - 3L1$$

$$= 3 - 3 \cdot 1 = 0$$

$$= 1 - 3 \cdot 1 = -2$$

$$= 1 - 3 \cdot 1 = -2$$

$$= 0 - 3 \cdot 0 = 0$$

|   |    |    |   |
|---|----|----|---|
| 1 | 1  | 1  | 0 |
| 0 | 0  | 0  | 0 |
| 0 | -2 | 0  | 0 |
| 0 | -2 | -2 | 0 |

Intercon L2 por L4

|   |    |    |   |
|---|----|----|---|
| 1 | 1  | 1  | 0 |
| 0 | -2 | -2 | 0 |
| 0 | -2 | 0  | 0 |
| 0 | 0  | 0  | 0 |

$$L3 \rightarrow L3 + L2$$

$$= -2 + -2 = 0$$

$$= 0 - (-2) = -2$$

$$= 0 + 0 = 0$$

| x | y  | z  |   |
|---|----|----|---|
| 1 | 1  | 1  | 0 |
| 0 | -2 | -2 | 0 |
| 0 | 0  | 2  | 0 |
| 0 | 0  | 0  | 0 |

Rank(A) = 3

Incognitas n = 3

Numero de Parametros = 3 - 3 = 0

Sistema Posiblemente Determinado SPD

$$\begin{cases} x + y + z = 0 \\ -2y - 2z = 0 \\ 0 - 2y = 0 \end{cases}$$

tilibra

Terceira Linha

$$-2y = 0$$

$$y = 0$$

$$-2$$

$$y = 0$$

$$-2y = 2y = 0$$

$$-2y - 2 \cdot 0 = 0$$

$$-2y - 0 = 0$$

$$y = 0$$

$$2$$

$$y = 0$$

$$x + y + z = 0$$

$$x + 0 + 0 = 0$$

$$x = 0$$

$$(x, y, z) = (0, 0, 0)$$

Lista de Exercícios

\* Resolva e classifique os sistemas.

$$\begin{cases} x + y = 5 \\ x - y = 1 \end{cases}$$

|   |    |   |
|---|----|---|
| 1 | 1  | 5 |
| 1 | -1 | 1 |

|   |   |
|---|---|
| 1 | L |
| 2 | L |
| m | m |

$$L2 \rightarrow L2 - L1$$

$$= 1 - 1 = 0$$

$$= -1 - 1 = -2$$

$$= 1 - 5 = -4$$

| x | y  |    |
|---|----|----|
| 1 | 1  | 5  |
| 0 | -2 | -4 |

$$\text{Rato} = 2$$

$$\begin{cases} x + y = 5 \\ -2y = -4 \end{cases}$$

$$-2y = -4$$

$$y = -2$$

$$-2$$

$$y = 2$$

$$x + y = 5$$

$$x + 2 = 5$$

$$x = 5 - 2$$

$$x = 3$$

$$(x, y) = (3, 2)$$

Sistema Possível Determinado (SPD)



|                                                        | x | y |   |
|--------------------------------------------------------|---|---|---|
| 2. $\begin{cases} 2x + y = 4 \\ x + y = 3 \end{cases}$ | ② | 1 | 4 |
|                                                        | 1 | 1 | 3 |

$$L2 \rightarrow 2L2 - L1$$

$$= 2 \cdot 1 - 2 = 0$$

$$= 2 \cdot 1 - 1 = 1$$

$$= 2 \cdot 3 - 4 = 2$$

|   |   |   |
|---|---|---|
| ② | 1 | 4 |
| 0 | ① | 2 |

$$\text{Rosto} = 2$$

|                                                  |             |              |
|--------------------------------------------------|-------------|--------------|
| $\begin{cases} 2x + y = 4 \\ 1y = 2 \end{cases}$ | $1y = 2$    | $2x + y = 4$ |
| $1y = 2$                                         | $y = 2 = 2$ | $2x + 2 = 4$ |
|                                                  | 1           | $2x = 4 - 2$ |

$$(x, y) = (1, 2)$$

$$2x = 2$$

$$x = 2 = 1$$

Sistema Possível Determinado (SPD)

$$2$$

|                                                         |   |   |   |
|---------------------------------------------------------|---|---|---|
| 3. $\begin{cases} x + 2y = 9 \\ 3x + y = 5 \end{cases}$ | ① | 2 | 9 |
|                                                         | 3 | 1 | 5 |

|                           | x | y |     |
|---------------------------|---|---|-----|
| L2 $\rightarrow$ L3 - 3L1 | ① | 2 | 9   |
| $= 3 - 3 \cdot 1 = 0$     | 0 | ⑤ | -16 |
| $= 9 - 3 \cdot 2 = -5$    |   |   |     |
| $= 5 - 3 \cdot 9 = -16$   |   |   |     |

|                                                          |             |                      |
|----------------------------------------------------------|-------------|----------------------|
| $\begin{cases} x + 2y = 9 \\ -5y = -16 \end{cases}$      | $-5y = -16$ | $x + 2y = 9$         |
| $-5y = -16$                                              | $y = -16$   | $x + 2 \cdot 16 = 9$ |
|                                                          | -5          | 5                    |
| $(x, y) = \begin{pmatrix} 3 & 16 \\ 5 & 5 \end{pmatrix}$ | $y = 16$    | $x + 32 = 9$         |
|                                                          | 5           | 5                    |

Sistema Possível Determinado

$$x = 9 - 32 = 35 - 32 = 3$$

|   |   |   |   |
|---|---|---|---|
| 1 | 5 | 5 | 5 |
|---|---|---|---|

tilibra

|                                                         |   |   |   |   |   |   |
|---------------------------------------------------------|---|---|---|---|---|---|
| 4. $\begin{cases} x + y = 2 \\ 2x + 2y = 4 \end{cases}$ | 1 | 1 | 2 | 1 | 2 | 2 |
|                                                         | 2 | 2 | 4 |   |   |   |

Rango = 1

|                           |   |   |   |  |  |  |
|---------------------------|---|---|---|--|--|--|
| $L2 \rightarrow L2 - 2L1$ | 0 | 0 | 0 |  |  |  |
| $= 2 - 2 \cdot 1 = 0$     |   |   |   |  |  |  |
| $= 2 - 2 \cdot 1 = 0$     |   |   |   |  |  |  |
| $= 4 - 2 \cdot 2 = 0$     |   |   |   |  |  |  |

$\begin{cases} x + y = 2 \\ x + y = 2 \end{cases}$

$x + y = 2$

$y = 2 - x$

$(x, y) = (x, 2 - x)$

Sistema Linear Indeterminado SPI

|                                                       |   |   |   |   |   |   |
|-------------------------------------------------------|---|---|---|---|---|---|
| 5. $\begin{cases} x + y = 3 \\ x + y = 5 \end{cases}$ | 1 | 1 | 3 | 1 | 1 | 5 |
|                                                       | 1 | 1 | 5 |   |   |   |

Rango = 1

|                          |   |   |   |  |  |  |
|--------------------------|---|---|---|--|--|--|
| $L2 \rightarrow L2 - L1$ | 0 | 0 | 2 |  |  |  |
| $= 1 - 1 = 0$            |   |   |   |  |  |  |
| $= 1 - 1 = 0$            |   |   |   |  |  |  |
| $= 5 - 3 = 2$            |   |   |   |  |  |  |

Sistema Impossível SI

|                                                               |   |    |   |   |   |   |
|---------------------------------------------------------------|---|----|---|---|---|---|
| 6. $\begin{cases} x + y + z = 3 \\ x - y + z = 1 \end{cases}$ | 1 | 1  | 1 | 3 | 1 | 1 |
|                                                               | 1 | -1 | 1 | 1 |   |   |

Rango = 2

|                          |   |    |   |    |  |  |
|--------------------------|---|----|---|----|--|--|
| $L2 \rightarrow L2 - L1$ | 0 | -2 | 0 | -2 |  |  |
| $= 1 - 1 = 0$            |   |    |   |    |  |  |
| $= -1 - 1 = -2$          |   |    |   |    |  |  |
| $= 1 - 1 = 0$            |   |    |   |    |  |  |
| $= 1 - 3 = -2$           |   |    |   |    |  |  |

$$\begin{cases} x + y + z = 3 \\ -2y = -2 \end{cases}$$

$$\begin{aligned} -2y &= -2 \\ y &= 1 \end{aligned}$$

$$\begin{aligned} x + y + z &= 3 \\ x + 1 + z &= 3 \\ x &= 3 - 1 - z \end{aligned}$$

$$\begin{aligned} y &= z \\ y &= 1 \\ x &= 2 - z \end{aligned}$$

$$y = 1$$

$$x = 2 - z$$

$$(x, y, z) = (2 - z, 1, z)$$

Sistema Posível Indeterminado (SPI)

$$\begin{cases} x + 2y + z = 4 \\ 2x + y + z = 5 \end{cases}$$

|   |   |   |   |
|---|---|---|---|
| 1 | 2 | 1 | 4 |
| 2 | 1 | 1 | 5 |

$$\begin{matrix} 1 & 2 \\ 3 & 1 \\ m & m \end{matrix}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 2 = -3$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 5 - 2 \cdot 4 = -3$$

| x | y | z  |    |
|---|---|----|----|
| ① | 2 | 1  | 4  |
| 0 | ③ | -1 | -3 |

$$\text{Rosto} = 2$$

$$\begin{cases} x + 2y + z = 4 \\ -3y + z = -3 \end{cases}$$

$$\begin{aligned} -3y + z &= -3 \\ z &= -3 + 3y \\ z &= 3 - 3y \end{aligned}$$

$$\begin{aligned} x + 2y + z &= 4 \\ x + 2y + (3 - 3y) &= 4 \\ x - y + 3 &= 4 \\ x &= 4 - 3 + y \\ x &= y + 1 \end{aligned}$$

$$(x, y, z) = (y + 1, y, 3 - 3y)$$

Sistema Posível Indeterminado (SPI)

$$2. \begin{cases} x + y + z = 2 \\ 2x + 2y + 2z = 4 \end{cases} \quad \begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 4 \end{array} \quad \begin{array}{c} 1 \quad 2 \\ 4 \quad 2 \\ N \quad M \end{array}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 4 - 2 \cdot 2 = 0$$

|   | X | y | z |   |
|---|---|---|---|---|
| ① | 1 | 1 | 1 | 2 |
|   | 0 | 0 | 0 | 0 |

Rank = 1

$$3. \begin{cases} x + y + z = 2 \end{cases}$$

$$y = y \quad x + y + z = 2$$

$$z = z \quad x = 2 - y - z$$

$$(x, y, z) = (2 - y - z, y, z)$$

Sistema Possível Indeterminado (SPI)

$$9. \begin{cases} x + y + z = 1 \\ x + y + z = 3 \end{cases} \quad \begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 3 \end{array} \quad \begin{array}{c} 1 \quad 2 \\ 3 \quad 3 \\ N \quad M \end{array}$$

$$L2 \rightarrow L2 - L1$$

$$= 1 - 1 = 0$$

$$= 1 - 1 = 0$$

$$= 1 - 1 = 0$$

$$= 3 - 1 = 2$$

|  | X | y | z |   |
|--|---|---|---|---|
|  | 1 | 1 | 1 | 1 |
|  | 0 | 0 | 0 | 2 |

Sistema Impossível (SI)

$$\begin{cases} x + y + z = 2 \\ 6x - y + z = 0 \end{cases}$$

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 6 & -1 & 1 & 0 \end{bmatrix}$$

$$\begin{matrix} 1 & L \\ 3 & 2 \\ m & m \end{matrix}$$

$$L2 \rightarrow L2 - L1$$

$$= 1 - 1 = 0$$

$$= -1 - 1 = -2$$

$$= 1 - 1 = 0$$

$$= 0 - 2 = -2$$

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & -2 & 0 & -2 \end{bmatrix}$$

$$\text{Rango} = 2$$

$$\begin{cases} x + y + z = 2 \\ -2y = -2 \end{cases}$$

$$-2y = -2$$

$$y = -2 / -2 = 1$$

$$-2$$

$$x + y + z = 2$$

$$x + 1 + z = 2$$

$$x = 2 - 1 - z$$

$$x = 1 - z$$

$$y = 1$$

$$(x, y, z) = (1 - z, 1, z)$$

Sistema Linear Indeterminado (SPI)

$$\begin{cases} x + y + z = 3 \\ 2x - y + z = 1 \\ 6x + 3y - z = 2 \end{cases}$$

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 1 & 1 \\ 6 & 3 & -1 & 2 \end{bmatrix}$$

$$\begin{matrix} 1 & L \\ 3 & 3 \\ N & M \end{matrix}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2(1) = 0$$

$$= -1 - 2(1) = -3$$

$$= 1 - 2(1) = -1$$

$$= 1 - 2(3) = -5$$

$$L3 \rightarrow L3 - L1$$

$$= 1 - 1 = 0$$

$$= 3 - 1 = 2$$

$$= -1 - 1 = -2$$

$$= 2 - 3 = -1$$

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -3 & -1 & -5 \\ 0 & 2 & -2 & -1 \end{bmatrix}$$

$$13 - 03 + 212$$

$$= 3 \cdot 2 + 2(-3) = 6 + 6 = 0$$

$$= 3(-2) + 2(-1) = -6 - 2 = -8$$

$$= 3(-1) + 2(-5) = -3 - 10 = -13$$

$$\begin{array}{ccc|c} \textcircled{1} & 1 & 1 & 3 \\ 0 & \textcircled{-3} & -1 & -5 \\ 0 & 0 & \textcircled{-8} & -13 \end{array}$$

$$\text{Rasio} = 3$$

$$\begin{cases} X + y + z = 3 \\ -3y - z = -5 \\ 0 - 8z = -13 \end{cases}$$

$$-8z = -13$$

$$-3y - z = -5$$

$$-3y - z = -5$$

$$z = -13$$

$$-3y - 13 = -5$$

$$0 - 8z = -13$$

$$-8$$

$$8$$

$$z = 13$$

$$-3y = -5 + 13$$

$$8$$

$$8$$

$$X + y + z = 3$$

$$X + 13 + 9 = 3$$

$$8 \quad 8$$

$$-3y = -40 + 13$$

$$8$$

$$X = 3 - 13 - 9$$

$$8 \quad 8$$

$$-3y = -29$$

$$8$$

$$X = 24 - 13 - 9$$

$$8$$

$$y = -29 : -3$$

$$8$$

$$X = 2^{12} = 1$$

$$8^{12} \quad 4$$

$$y = -29 \cdot 1$$

$$8 \quad 3$$

$$y = -29^{13} \quad 9$$

$$24^{13} \quad 8$$

$$(X, y, z) = \left(\frac{1}{4}, \frac{9}{8}, \frac{13}{8}\right)$$

Sistematik Penyelesaian Determinan (SPD)

$$\begin{cases} 2X + y + z = 4 \\ X - y + z = 1 \\ X + 2y - z = 3 \end{cases}$$

$$\begin{array}{ccc|c} \textcircled{2} & 1 & 1 & 4 \\ 1 & -1 & 1 & 1 \\ 1 & 2 & -1 & 3 \end{array}$$

$$\begin{array}{c} 1 \quad 2 \\ 3 \quad 3 \\ N \quad M \end{array}$$

$$L2 \rightarrow L2 - L1$$

$$= 2(1) - 2 = 0$$

$$= 2(-1) - 1 = -3$$

$$= 2(1) - 1 = 1$$

$$= 2(1) - 4 = -2$$

$$L3 \rightarrow L3 - L1$$

$$= 2(1) - 2 = 0$$

$$= 2(2) - 1 = 3$$

$$= 2(-1) - 1 = -3$$

$$= 2(3) - 4 = 2$$

|   |    |    |    |
|---|----|----|----|
| 2 | 1  | 1  | 4  |
| 0 | -3 | 1  | -2 |
| 0 | 3  | -3 | 2  |

$$L3 \rightarrow L3 + L2$$

$$= 3 + (-3) = 0$$

$$= -3 + 1 = -2$$

$$= 2 + (-2) = 0$$

| x | y  | z |    |
|---|----|---|----|
| 2 | 1  | 1 | 4  |
| 0 | -3 | 1 | -2 |
| 0 | 0  | 2 | 0  |

Rank = 3

$$\begin{cases} 2x + y + z = 4 \\ -3y + z = -2 \\ -2z = 0 \end{cases}$$

$$-2z = 0$$

$$z = 0 = 0$$

$$-3y + z = -2$$

$$-3y + 0 = -2$$

$$-3y = -2$$

$$y = \frac{-2}{-3} = \frac{2}{3}$$

$$2x + y + z = 4$$

$$2x + \frac{2}{3} + 0 = 4$$

$$(x, y, z) = \left(\frac{5}{3}, \frac{2}{3}, 0\right)$$

$$2x = 4 - 2$$

Sistema Positivo Determinado (SPD)

$$2x = 12 - 2$$

$$2x = 10$$

$$x = 10 : 2$$

$$x = \frac{10}{2} = 5$$

$$\frac{10}{2} = 5$$

\* 13. 
$$\begin{cases} x + y + z = 1 \\ 2x + 2y + 2z = 2 \\ x - y + z = 0 \end{cases}$$

|   |   |    |   |   |
|---|---|----|---|---|
| ① | 1 | 1  | 1 | 1 |
| 2 | 2 | 2  | 2 | 2 |
| 3 | 1 | -1 | 1 | 0 |

$$\begin{array}{cc} 1 & L \\ 3 & 3 \\ M & M \end{array}$$

$L2 \rightarrow L2 - 2L1$

$= 2 - 2(1) = 0$

$= 2 - 2(1) = 0$

$= 2 - 2(1) = 0$

$= 2 - 2(1) = 0$

$L3 \rightarrow L3 - L2$

$= 1 - 1 = 0$

$= -1 - 1 = -2$

$= 1 - 1 = 0$

$= 0 - 1 = -1$

|   |    |   |    |   |
|---|----|---|----|---|
| ① | 1  | 1 | 1  | 1 |
| 0 | 0  | 0 | 0  | 0 |
| 0 | -2 | 0 | -1 | 0 |

Interacar L3 por L2

|   |    |   |    |   |
|---|----|---|----|---|
| ① | 1  | 1 | 1  | 1 |
| 0 | -2 | 0 | -1 | 0 |
| 0 | 0  | 0 | 0  | 0 |

Paso 2  $u = y + z + x$

$$\begin{cases} x + y + z = 1 \\ -2y = -1 \end{cases}$$

$-2y = -1$

$y = \frac{1}{2}$

2

$z = t$

$x = 1 - \frac{1}{2} - t$

2

$(x, y, z) = \left(\frac{1}{2} - t, \frac{1}{2}, t\right)$

$x = \frac{1}{2} - t - t$

2

Sistema Posiblemente Indeterminado (SPI)

$x = \frac{1}{2} - t$

2

\* 14. 
$$\begin{cases} x + y + z = 1 \\ 2x + 2y + 2z = 3 \\ x - y + z = 0 \end{cases}$$

|   |   |    |   |
|---|---|----|---|
| 1 | 1 | 1  | 1 |
| 2 | 2 | 2  | 3 |
| 3 | 1 | -1 | 0 |



|                           |                          |                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |    |   |   |
|---------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|----|---|---|
| $L2 \rightarrow L2 - 2L1$ | $L3 \rightarrow L3 - L1$ | <table border="1"> <tr><td>1</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>0</td><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>-2</td><td>0</td><td>1</td></tr> </table> | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | -2 | 0 | 1 |
| 1                         | 1                        | 1                                                                                                                                                                                  | 1 |   |   |   |   |   |   |   |   |    |   |   |
| 0                         | 0                        | 0                                                                                                                                                                                  | 1 |   |   |   |   |   |   |   |   |    |   |   |
| 0                         | -2                       | 0                                                                                                                                                                                  | 1 |   |   |   |   |   |   |   |   |    |   |   |
| $= 2 - 2 \cdot 1 = 0$     | $= 1 - 1 = 0$            |                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |    |   |   |
| $= 2 - 2 \cdot 1 = 0$     | $= -1 - 1 = -2$          |                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |    |   |   |
| $= 2 - 2 \cdot 1 = 0$     | $= 1 - 1 = 0$            |                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |    |   |   |
| $= 3 - 2 \cdot 1 = 1$     | $= 0 - 1 = -1$           |                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |    |   |   |

Sistema Impossível pois  $0 = 1$

\*15.  $\begin{cases} x + y + z = 2 \\ x + 2y + z = 3 \\ x + y + 2z = 4 \end{cases}$

|   |   |   |   |
|---|---|---|---|
| ① | 1 | 1 | 2 |
|   | 1 | 2 | 3 |
|   | 1 | 1 | 4 |

$\begin{matrix} I & L \\ 3 & 3 \\ N & M \end{matrix}$

|                          |                          |                                                                                                                                                                                 |   |   |   |   |  |   |   |   |  |   |   |   |          |
|--------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|--|---|---|---|--|---|---|---|----------|
| $L2 \rightarrow L2 - L1$ | $L3 \rightarrow L3 - L1$ | <table border="1"> <tr><td>①</td><td>1</td><td>1</td><td>2</td></tr> <tr><td></td><td>0</td><td>①</td><td>1</td></tr> <tr><td></td><td>0</td><td>0</td><td>②</td></tr> </table> | ① | 1 | 1 | 2 |  | 0 | ① | 1 |  | 0 | 0 | ② | Rank = 3 |
| ①                        | 1                        | 1                                                                                                                                                                               | 2 |   |   |   |  |   |   |   |  |   |   |   |          |
|                          | 0                        | ①                                                                                                                                                                               | 1 |   |   |   |  |   |   |   |  |   |   |   |          |
|                          | 0                        | 0                                                                                                                                                                               | ② |   |   |   |  |   |   |   |  |   |   |   |          |
| $= 1 - 1 = 0$            | $= 1 - 1 = 0$            |                                                                                                                                                                                 |   |   |   |   |  |   |   |   |  |   |   |   |          |
| $= 2 - 1 = 1$            | $= 1 - 1 = 0$            |                                                                                                                                                                                 |   |   |   |   |  |   |   |   |  |   |   |   |          |
| $= 1 - 1 = 0$            | $= 2 - 1 = 1$            |                                                                                                                                                                                 |   |   |   |   |  |   |   |   |  |   |   |   |          |
| $= 3 - 2 = 1$            | $= 4 - 2 = 2$            |                                                                                                                                                                                 |   |   |   |   |  |   |   |   |  |   |   |   |          |

|                                                             |                                                                              |
|-------------------------------------------------------------|------------------------------------------------------------------------------|
| $\begin{cases} x + y + z = 2 \\ y = 1 \\ z = 2 \end{cases}$ | $x + y + z = 2$<br>$x + 1 + 2 = 2$<br>$x + 3 = 2$<br>$x = 2 - 3$<br>$x = -1$ |
|-------------------------------------------------------------|------------------------------------------------------------------------------|

$(x, y, z) = (-1, 1, 2)$

Sistema Possível Determinado (SPD)

\*16. 
$$\begin{cases} 2x + y + z = 5 \\ x + y + z = 3 \\ x - y + z = 1 \end{cases}$$

|   |    |   |   |
|---|----|---|---|
| 2 | 1  | 1 | 5 |
| 1 | 1  | 1 | 3 |
| 1 | -1 | 1 | 1 |

$$\begin{matrix} 1 & 1 \\ 3 & 3 \\ m & m \end{matrix}$$

L2 → L2 - L1

$= 2 \cdot 1 - 2 = 0$

$= 2 \cdot 1 - 1 = 1$

$= 2 \cdot 1 - 1 = 1$

$= 2 \cdot 3 - 5 = 1$

L3 = 2L3 - L1

$= 2 \cdot 1 - 2 = 0$

$= 2 \cdot 1 - 1 = -3$

$= 2 \cdot 1 - 1 = 1$

$= 2 \cdot 3 - 5 = -3$

|   |    |   |    |
|---|----|---|----|
| 2 | 1  | 1 | 5  |
| 0 | 1  | 1 | 1  |
| 0 | -3 | 1 | -3 |

L3 → L3 + 3L2

$= -3 + 3 \cdot 1 = 0$

$= 1 + 3 \cdot 1 = 4$

$= -3 + 3 \cdot 1 = 0$

|   |   |   |   |
|---|---|---|---|
| 2 | 1 | 1 | 5 |
| 0 | 1 | 1 | 1 |
| 0 | 0 | 4 | 0 |

Rank = 3

$$\begin{cases} 2x + y + z = 5 \\ y + z = 1 \\ 4z = 0 \end{cases}$$

$4z = 0$

$z = 0 = 0$

$4$

$y + z = 1$

$y + 0 = 1$

$y = 1 - 0$

$y = 1$

$2x + y + z = 5$

$2x + 1 + 0 = 5$

$2x = 5 - 1$

$2x = 4$

$x = 4 / 2 = 2$

$(x, y, z) = (2, 1, 0)$

Sistema Resolvido Determinado (SPD)

\*19. 
$$\begin{cases} x + 2y + z = 1 \\ 2x + 4y + 2z = 2 \\ 3x + 6y + 3z = 3 \end{cases}$$

|   |   |   |   |
|---|---|---|---|
| 1 | 2 | 1 | 1 |
| 2 | 4 | 2 | 2 |
| 3 | 6 | 3 | 3 |

$$\begin{matrix} 1 & 1 \\ 3 & 3 \\ m & m \end{matrix}$$



$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 4 - 2 \cdot 2 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$L3 \rightarrow L3 - 3L1$$

$$= 3 - 3 \cdot 1 = 0$$

$$= 6 - 3 \cdot 2 = 0$$

$$= 3 - 3 \cdot 1 = 0$$

$$= 3 - 3 \cdot 1 = 0$$

$$\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \quad \text{Rosto} = 1$$

$$\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}$$

$$\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}$$

$$\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}$$

$$\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}$$

$$\begin{cases} x + 2y + z = 1 \end{cases}$$

$$x + 2y + z = 1$$

$$x = 1 - 2y - z$$

$$(x, y, z) = (1 - 2y - z, y, z)$$

Sistema Posível Indeterminado (SPI)

$$\begin{cases} x + y + z + t = 2 \\ 2x + y + z + t = 3 \end{cases}$$

$$\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 2 & 1 & 1 & 1 & 3 \end{array}$$

$$\begin{array}{cc} 1 & 2 \\ 2 & 3 \\ N & M \end{array}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 3 - 2 \cdot 2 = -1$$

$$\begin{array}{ccc|c} 1 & -1 & -1 & 2 \\ 0 & -1 & -1 & -1 \end{array} \quad \text{Rosto} = 2$$

$$\begin{array}{ccc|c} 0 & -1 & -1 & -1 \\ 0 & -1 & -1 & -1 \end{array}$$

$$\begin{array}{ccc|c} 0 & -1 & -1 & -1 \\ 0 & -1 & -1 & -1 \end{array}$$

$$\begin{array}{ccc|c} 0 & -1 & -1 & -1 \\ 0 & -1 & -1 & -1 \end{array}$$

$$\begin{cases} x + y - z - t = 2 \end{cases}$$

$$-y - z - t = -1$$

$$-y = -1 + z + t$$

$$y = 1 - z - t$$

$$x + y - z - t = 2 + y - y - x$$

$$x + (1 - z - t) - z - t = 2$$

$$x = 2 - 1$$

$$t = t$$

$$x = 1$$

$$z = z$$

$$(x, y, z, t) = (1, 1 - y - z - t, y, z)$$

Sistema Posível Indeterminado SPI

$$19. \begin{cases} x + y + z + t = 1 \\ x - y + z + t = 0 \end{cases} \quad \left[ \begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & 0 \end{array} \right] \quad \begin{matrix} 1 & L \\ 4 & J \\ N & M \end{matrix}$$

$$12 - 12 - 11$$

$$= 1 - 1 = 0$$

$$= -1 - 1 = -2$$

$$= 1 - 1 = 0$$

$$= 1 - 1 = 0$$

$$= 0 - 1 = -1$$

$$\left[ \begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & -2 & 0 & 0 & -1 \end{array} \right] \quad \text{Rosto} = 2$$

$$\begin{cases} x + y + z + t = 1 \\ -2y = -1 \end{cases} \quad \begin{matrix} -2y = -1 \\ y = \frac{1}{2} \end{matrix} \quad \begin{matrix} x + y + z + t = 1 \\ x + \frac{1}{2} + z + t = 1 \end{matrix} \quad \begin{matrix} 2 & 1 \\ 1 & 2 \end{matrix}$$

$$y = y$$

$$t = t$$

$$x = 1 - \frac{1}{2} - y - t$$

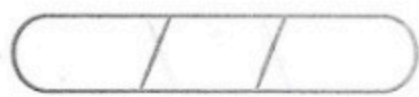
$$x = 2 - 1 - y - t$$

$$(x, y, z, t) = \left( \frac{1}{2} - y - t, \frac{1}{2}, y, t \right)$$

$$x = 1 - y - t$$

Sistema Impossível Indeterminado (SPI)

$$20. \begin{cases} x + y + z + t = 2 \\ 2x + 2y + 2z + 2t = 4 \end{cases} \quad \left[ \begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{array} \right] \quad \begin{matrix} 1 & L \\ 4 & J \\ 2m & m \end{matrix}$$



|                           |                   |   |   |   |   |           |
|---------------------------|-------------------|---|---|---|---|-----------|
| $L2 \rightarrow L2 - 2L1$ | $\textcircled{1}$ | 4 | 1 | 1 | 2 | Posto = 1 |
| $= 2 - 2(1) = 0$          | 0                 | 0 | 0 | 0 | 0 |           |
| $= 2 - 2(1) = 0$          |                   |   |   |   |   |           |
| $= 2 - 2(1) = 0$          |                   |   |   |   |   |           |
| $= 2 - 2(1) = 0$          |                   |   |   |   |   |           |
| $= 4 - 2(2) = 0$          |                   |   |   |   |   |           |

$$X + y + z + t = 2$$

$$X + y + z + t = 2$$

$$X = 2 - y - z - t$$

$$(X, y, z, t) = (2 - y - z - t, y, z, t)$$

21. Determine o posto

|                                                |                           |                   |   |           |
|------------------------------------------------|---------------------------|-------------------|---|-----------|
| $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ | $L2 \rightarrow L2 - 2L1$ | $\textcircled{1}$ | 2 | Posto = 1 |
|                                                | $= 2 - 2 \cdot 1 = 0$     | 0                 | 0 |           |
|                                                | $= 4 - 2 \cdot 2 = 0$     |                   |   |           |

22. Determine o posto

|                                                        |                           |                   |                   |   |           |
|--------------------------------------------------------|---------------------------|-------------------|-------------------|---|-----------|
| $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ | $L2 \rightarrow L2 - 2L1$ | $\textcircled{1}$ | 0                 | 1 | Posto = 2 |
|                                                        | $= 2 - 2 \cdot 1 = 0$     | 0                 | $\textcircled{1}$ | 1 |           |
|                                                        | $= 1 - 2 \cdot 0 = 1$     |                   |                   |   |           |
|                                                        | $= 3 - 2 \cdot 1 = 1$     |                   |                   |   |           |

23. Determine o posto

|                                                                     |                           |                          |
|---------------------------------------------------------------------|---------------------------|--------------------------|
|                                                                     | $L2 \rightarrow L2 - 2L1$ | $L3 \rightarrow L3 - L1$ |
|                                                                     | $= 2 - 2 \cdot 1 = 0$     | $= 1 - 1 = 0$            |
|                                                                     | $= 4 - 2 \cdot 2 = 0$     | $= 1 - 2 = -1$           |
|                                                                     | $= 6 - 2 \cdot 3 = 0$     | $= 1 - 3 = -2$           |
| $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$ |                           |                          |

tilibra

|   |    |    |                    |   |    |    |           |
|---|----|----|--------------------|---|----|----|-----------|
| 1 | 2  | 3  | Subtrair L2 por L3 | 1 | 2  | 3  | Ranko = 2 |
| 0 | 0  | 0  |                    | 0 | -1 | -2 |           |
| 0 | -1 | -2 |                    | 0 | 0  | 0  |           |

24. Resolver

$$\begin{cases} x + y + z = 0 \\ x - y + z = 0 \end{cases} \quad \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{array} \quad \begin{array}{cc} 1 & L \\ 3 & 2 \\ m & m \end{array}$$

$$\begin{aligned} L2 \rightarrow L2 - L1 \\ = 1 - 1 = 0 \\ = -1 - 1 = -2 \\ = 1 - 1 = 0 \\ = 0 - 0 = 0 \end{aligned} \quad \begin{array}{ccc|c} x & y & z & \\ 1 & 1 & 1 & 0 \\ 0 & -2 & 0 & 0 \end{array} \quad \text{Ranko} = 2$$

$$\begin{cases} x + y + z = 0 \\ -2y = 0 \end{cases} \quad \begin{aligned} -2y &= 0 \\ y &= 0 \end{aligned} \quad \begin{aligned} x + y + z &= 0 \\ x + 0 + z &= 0 \\ x &= -z \end{aligned}$$

$$z = t$$

$$(x, y, z) = (-t, 0, t)$$

Sistema Possível Indeterminado (SPI)

25. Resolver

$$\begin{cases} x + y + z = 0 \\ 2x + 2y + 2z = 0 \end{cases} \quad \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 \end{array} \quad \begin{array}{cc} 1 & L \\ 3 & 2 \\ m & m \end{array}$$



$$L2 \rightarrow L2 - 2L1$$

|   |   |   |   |
|---|---|---|---|
| ① | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

$$\text{Rango} = 1$$

$$= 2 - 2(1) = 0$$

$$= 2 - 2(1) = 0$$

$$= 2 - 2(1) = 0$$

$$= 0 - 2(0) = 0$$

$$\begin{cases} x + y + z = 0 \\ x + y + z = 0 \end{cases}$$

$$x = -y - z$$

$(x, y, z) = (-y - z, y, z)$   
 Sistema Positivo Indeterminado (SPI)

26. Resolver

$$\begin{cases} x + y + z + t = 0 \\ x - y + z + t = 0 \end{cases}$$

|   |    |   |   |   |
|---|----|---|---|---|
| 1 | 1  | 1 | 1 | 0 |
| 1 | -1 | 1 | 1 | 0 |

$$\begin{matrix} 1 & 2 \\ 4 & 2 \\ m & m \end{matrix}$$

$$L2 \rightarrow L2 - L1$$

|   |    |   |   |   |
|---|----|---|---|---|
| ① | 1  | 1 | 1 | 0 |
| 0 | -2 | 0 | 0 | 0 |

$$\text{Rango} = 2$$

$$= 1 - 1 = 0$$

$$= -1 - 1 = -2$$

$$= 1 - 1 = 0$$

$$= 1 - 1 = 0$$

$$= 0 - 0 = 0$$

$$\begin{cases} x + y + z + t = 0 \\ -y = 0 \end{cases}$$

$$-y = 0$$

$$y = 0$$

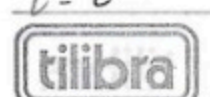
$$x + y + z + t = 0$$

$$x + 0 + z + t = 0$$

$$x = -z - t$$

$$y = 0$$

$$z = z$$



$$(x, y, z, t) = (-y - z, 0, y, z)$$

Sistema Linear Indeterminado (SLI)

27. Classificar

$$\begin{cases} x + y + z = 4 \\ 2x + 2y + 2z = 8 \end{cases} \quad \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 2 & 2 & 8 \end{array} \right] \quad \begin{array}{c} 1 \quad 2 \\ 3 \quad 2 \\ m \quad m \end{array}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 8 - 2 \cdot 4 = 0$$

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\text{Rango} = 1$$

$$\begin{cases} x + y + z = 4 \end{cases}$$

$$x + y + z = 4$$

$$x = 4 - y - z$$

$$y = y$$

$$z = z$$

$$(x, y, z) = (-y - z, y, z)$$

Sistema Linear Indeterminado (SLI)

28. Classificar

$$\begin{cases} x + y + z = 4 \\ 2x + 2y + 2z = 8 \end{cases} \quad \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 2 & 2 & 8 \end{array} \right] \quad \begin{array}{c} 1 \quad 2 \\ 3 \quad 2 \\ m \quad m \end{array}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 4 - 2 \cdot 4 = -4$$

$$\begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$\text{Rosto} = 1$$

Sistema impossível pelo fato que deu  $0 = -1$

29. Resolver

$$\begin{cases} x + y + z = 3 \\ 2x + y + z = 4 \\ x - y + z = 2 \end{cases}$$

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & 1 & 1 & 4 \\ 1 & -1 & 1 & 2 \end{bmatrix}$$

$$\begin{matrix} 1 & L \\ 3 & 3 \\ m & m \end{matrix}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 4 - 2 \cdot 3 = -2$$

$$L3 \rightarrow L3 - L1$$

$$= 1 - 1 = 0$$

$$= -1 - 1 = -2$$

$$= 1 - 1 = 0$$

$$= 2 - 3 = -1$$

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -1 & -1 & -2 \\ 0 & -2 & 0 & -1 \end{bmatrix}$$

$$L3 \rightarrow L3 - 2L2$$

$$= -2 - 2(-1) = 0$$

$$= 0 - 2(-1) = 2$$

$$= -1 - 2(-2) = 3$$

$$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -1 & -1 & -2 \\ 0 & 0 & 2 & 3 \end{bmatrix}$$

$$\text{Rosto} = 3$$

$$\begin{cases} x + y + z = 3 \\ -y - z = -2 \\ -2z = 3 \end{cases}$$

$$-2y = 3$$

$$y = \frac{3}{2}$$

$$-y - z = -2$$

$$-y - 3 = -2$$

$$\frac{-y}{2} = \frac{-2+3}{2}$$

$$-y = -2 + 3$$

$$\frac{-y}{2} = \frac{-2+3}{2}$$

$$-y = -4 + 3$$

$$\frac{-y}{2} = \frac{-4+3}{2}$$

$$-y = -1$$

$$\frac{-y}{2} = \frac{-1}{2}$$

$$y = \frac{1}{2}$$

$$\frac{y}{2} = \frac{1}{2}$$

$$x + y + z = 3$$

$$x + \frac{1}{2} + \frac{3}{2} = 0$$

$$\frac{x}{2} = \frac{0 - \frac{1}{2} - \frac{3}{2}}{2}$$

$$x + \frac{1}{2} = 3$$

$$\frac{x}{2} = \frac{3 - \frac{1}{2}}{2}$$

$$x = -4 + 3$$

$$\frac{x}{2} = \frac{-4+3}{2}$$

$$x = -4 + 6 = 2 = 1$$

$$\frac{x}{2} = \frac{-4+6}{2} = \frac{2}{2} = 1$$

$$(x, y, z) = \left(1, \frac{1}{2}, \frac{3}{2}\right)$$

Sistema Linear Determinado (SLD)

30. Resolver

$$\begin{cases} x + y + z = 1 \\ x + 2y + z = 2 \\ x + y + 2z = 3 \end{cases}$$

$$\begin{cases} x + 2y + z = 2 \\ x + y + 2z = 3 \end{cases}$$

$$\begin{cases} x + y + 2z = 3 \end{cases}$$

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 1 | 1 |
| 1 | 2 | 1 | 2 |
| 1 | 1 | 2 | 3 |

1 2  
3 3  
N M

$$L2 \rightarrow L2 - L1$$

$$= 1 - 1 = 0$$

$$= 2 - 1 = 1$$

$$= 1 - 1 = 0$$

$$= 2 - 1 = 1$$

$$L3 \rightarrow L3 - L1$$

$$= 1 - 1 = 0$$

$$= 1 - 1 = 0$$

$$= 2 - 1 = 1$$

$$= 3 - 1 = 2$$

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 2 |

Rank = 3

$$\begin{cases} x+y+z=1 \\ y=1 \\ z=2 \end{cases} \quad \begin{aligned} x+y+z &= 1 \\ x+1+2 &= 1 \\ x &= 1-3 \\ x &= -2 \end{aligned}$$

$$(x, y, z) = (-2, 1, 2)$$

Sistema Possível Determinado (SPD)

\* Resolva e classifique os sistemas homogêneos abaixo:

$$31. \begin{cases} x+y+z=0 \\ x-y+z=0 \\ 2x+2y+2z=0 \end{cases} \quad \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & 0 \\ 2 & 2 & 2 & 0 \end{array} \right] \quad \begin{array}{cc} J & L \\ 3 & 3 \\ m & m \end{array}$$

$$\begin{array}{ll} L2 \rightarrow L2 - L1 & L3 \rightarrow L3 - 2L1 \\ = 1 - 1 = 0 & = 2 - 2 \cdot 1 = 0 \\ = -1 - 1 = -2 & = 2 - 2 \cdot 1 = 0 \\ = 1 - 1 = 0 & = 2 - 2 \cdot 1 = 0 \\ = 0 - 0 = 0 & = 0 - 2 \cdot 0 = 0 \end{array} \quad \begin{array}{ccc|c} x & y & z & \\ \hline 1 & 1 & 1 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}$$

$$\text{Rango } r(A) = 2$$

$$\text{Incertas } m = 3$$

$$\text{Número de Parâmetros} = 3 - 2 = 1$$

Sistema Possível Indeterminado (SPI)

$$\begin{cases} x+y+z=0 \\ -2y=0 \end{cases} \quad \begin{aligned} -2y &= 0 \\ y &= 0 \end{aligned} \quad \begin{aligned} x+y+z &= 0 \\ x+0+z &= 0 \\ x &= -z \end{aligned}$$

$$(x, y, z) = (-z, 0, z)$$

$$z(-1, 0, 1)$$



32. 
$$\begin{cases} x + y + z = 0 \\ 2x + y + z = 0 \\ 3x + 2y + 3z = 0 \end{cases}$$

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 1 | 0 |
| 2 | 1 | 1 | 0 |
| 3 | 2 | 3 | 0 |

$$\begin{array}{c} 1 \quad 2 \\ 3 \quad 3 \\ \hline m \quad m \end{array}$$

$L2 \rightarrow L2 - 2L1$

$= 2 - 2 \cdot 1 = 0$

$= 1 - 2 \cdot 1 = -1$

$= 1 - 2 \cdot 1 = -1$

$= 0 - 2 \cdot 0 = 0$

$L3 \rightarrow L3 - L1$

$= 1 - 1 = 0$

$= 2 - 1 = 1$

$= 3 - 1 = 2$

$= 0 - 0 = 0$

|   |    |    |   |
|---|----|----|---|
| 1 | 1  | 1  | 0 |
| 0 | -1 | -1 | 0 |
| 0 | 1  | 2  | 0 |

$L3 \rightarrow L3 + L2$

$= 1 + (-1) = 0$

$= 2 + (-1) = 1$

$= 0 + 0 = 0$

|   |    |    |   |
|---|----|----|---|
| 1 | 1  | 1  | 0 |
| 0 | -1 | -1 | 0 |
| 0 | 0  | 1  | 0 |

Rosto  $|A| = 3$

Incertas  $m = 3$

Numero de Parametros  $= 3 - 3 = 0$

Sistema Positivo Determinado (SPD)

$$\begin{cases} x + y + z = 0 \\ -y - z = 0 \\ -z = 0 \end{cases}$$

$-y - z = 0$

$-y - 0 = 0$

$-y = 0$

$x + y + z = 0$

$x + 0 + 0 = 0$

$x = 0$

$z = 0$

$(x, y, z) = (0, 0, 0)$

33. 
$$\begin{cases} x + y + z + w = 0 \\ 2x + y + z + w = 0 \\ x - y + z + w = 0 \end{cases}$$

|   |    |   |   |   |
|---|----|---|---|---|
| 1 | 1  | 1 | 1 | 0 |
| 2 | 1  | 1 | 1 | 0 |
| 1 | -1 | 1 | 1 | 0 |

$$\begin{array}{c} 1 \quad 2 \\ 4 \quad 3 \\ \hline m \quad m \end{array}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 1 - 2 \cdot 1 = -1$$

$$= 0 - 2 \cdot 0 = 0$$

$$L3 = L3 - L1$$

$$= 1 - 1 = 0$$

$$= -1 - 1 = -2$$

$$= 1 - 1 = 0$$

$$= 1 - 1 = 0$$

$$= 0 - 0 = 0$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & -1 & -1 & 0 \\ 0 & -2 & 0 & 0 & 0 \end{bmatrix}$$

$$0 = x + y + z + t$$

$$0 = -x - y - z - t$$

$$0 = 2x + 2y + 2z + 2t$$

$$L3 \rightarrow L3 - 2L2$$

$$= -2 - 2(-1) = 0$$

$$= 0 - 2(1) = -2$$

$$= 0 - 2(-1) = 2$$

$$= 0 - 2 \cdot 0 = 0$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & -1 & -1 & 0 \\ 0 & 0 & 2 & 2 & 0 \end{bmatrix}$$

$$\text{Rango } r(A) = 3$$

$$\text{Incógnitas } m = 4$$

$$\text{Número de Parâmetros} = 4 - 3 = 1$$

Sistema Possível Indeterminado (SPI)

$$\begin{cases} x + y + z + t = 0 \\ -y - z - t = 0 \\ 2z + 2t = 0 \end{cases}$$

$$2z + 2t = 0$$

$$2z = -2t$$

$$z = -t$$

$$z = -t$$

$$-y - z - t = 0$$

$$-y - (-t) - t = 0$$

$$-y + t - t = 0$$

$$-y = 0 - t + t$$

$$y = 0$$

$$x + y + z + t = 0$$

$$x + 0 + (-t) + t = 0$$

$$x - t + t = 0$$

$$x = 0$$

$$t = t$$

$$z = -t$$

$$(x, y, z, t) = (0, 0, -t, t)$$

$$= t(0, 0, -1, 1)$$

$$34. \begin{cases} x + y + z + t = 0 \\ 2x + 2y + 2z + 2t = 0 \end{cases}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 2 & 0 \end{bmatrix}$$

$$\begin{matrix} 1 & L \\ 2 & 2 \\ m & m \end{matrix}$$

$$L2 \rightarrow L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 0 - 2 \cdot 0 = 0$$

$$\begin{array}{ccc|c} \textcircled{1} & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}$$

$$\text{Rosto } n(A) = 1$$

$$\text{Incógnitas} = 4$$

$$\text{Número de Parâmetros} = 4 - 1 = 3$$

Sistema Possível Indeterminado (SPI)

$$x + y + z + t = 0$$

$$x + y + z + t = 0$$

$$x + z + t + t = 0$$

$$y = t$$

$$x = t - z - t$$

$$y = t$$

$$z = t$$

$$(x, y, z, t) = (t - t - t, t, t, t)$$

$$(1 - 1 - 1, 1, 1, 1)$$

$$\begin{array}{l} 35. \left\{ \begin{array}{l} x + y + z = 0 \\ 2x + 2y + 2z = 0 \\ x - y + z = 0 \\ 3x + y + z = 0 \end{array} \right. \end{array}$$

$$\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 \\ 1 & -1 & 1 & 0 \\ 3 & 1 & 1 & 0 \end{array}$$

$$\begin{array}{l} 1 \quad 2 \\ 3 \quad 4 \\ m \quad m \end{array}$$

$$L2 = L2 - 2L1$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 2 - 2 \cdot 1 = 0$$

$$= 0 - 2 \cdot 0 = 0$$

$$L3 = L3 - L1$$

$$= 1 - 1 = 0$$

$$= -1 - 1 = -2$$

$$= 1 - 1 = 0$$

$$= 0 - 0 = 0$$

$$L4 = L4 - 3L1$$

$$= 3 - 3 \cdot 1 = 0$$

$$= 1 - 3 \cdot 1 = -2$$

$$= 1 - 3 \cdot 1 = -2$$

$$= 0 - 3 \cdot 0 = 0$$

tilibra

|   |    |    |   |                     |   |    |    |   |
|---|----|----|---|---------------------|---|----|----|---|
| 1 | 1  | 1  | 0 | Substitua L2 com L4 | 1 | 1  | 1  | 0 |
| 0 | 0  | 0  | 0 |                     | 0 | -2 | -2 | 0 |
| 0 | -2 | 0  | 0 |                     | 0 | -2 | 0  | 0 |
| 0 | -2 | -2 | 0 |                     | 0 | 0  | 0  | 0 |

|                   |     |      |     |   |
|-------------------|-----|------|-----|---|
| $L3 = L3 - L2$    | (1) | 1    | 1   | 0 |
| $= -2 - (-2) = 0$ | 0   | (-2) | -2  | 0 |
| $= 0 - (-2) = 2$  | 0   | 0    | (2) | 0 |
| $= 0 - 0 = 0$     | 0   | 0    | 0   | 0 |

Rank  $r(A) = 3$

Incógnitas = 3

Numero de Parâmetros =  $3 - 3 = 0$

Sistema Possível Determinado (SPD)

$$\begin{cases} X + y + z = 0 \\ -2y + 2z = 0 \\ 2z = 0 \end{cases} \quad \begin{cases} 2z = 0 \\ z = 0 \\ z = 0 \end{cases} \quad \begin{cases} -2y + 2z = 0 \\ -2y + 2,0 = 0 \\ -2y + 0 = 0 \\ -2y = 0 \\ y = 0 \end{cases} \quad \begin{cases} X + y + z = 0 \\ X + 0 + 0 = 0 \\ X = 0 \end{cases}$$

$(X + y + z) = (0, 0, 0)$